

6. The following extracts are adapted from articles published on the Internet.

Hidden ocean found on Saturn's icy moon, Enceladus, could potentially support life.

Enceladus harbours a big ocean of liquid water beneath its icy crust that may be capable of supporting life as we know it. Researchers said that the water is about 10 kilometres deep and lies beneath a shell of ice 30 to 40 km thick. Furthermore, it is in direct contact with a rocky sea floor, theoretically making possible all kinds of complex chemical reactions – such as the kind that may have led to the rise of life on Earth.

NASA finds more evidence that the ocean on Enceladus could support alien life.

NASA's Cassini spacecraft orbiting the moon has analysed the plumes of gas forced out through fissures in the ice. These plumes have been found to contain four of the six most important elements of life on Earth – carbon, hydrogen, nitrogen and oxygen – only phosphorus and sulfur have not been detected.

Describe the structure of water and explain how the properties of water would be essential to supporting life on Enceladus. Explain how the absence of phosphorus and sulfur would prevent the formation of biochemical molecules essential for life on Earth. [9 QER]

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